

IRELAND WEST AP KNOCK, Ireland

WMO: 039730

Lat: 53.910N

Lon: 8.818W

Elev: 665

StdP: 14.35

Time Zone: 1.00 (IRL)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	28.1	30.2	23.7	18.4	31.7	26.0	20.5	33.3	34.3	46.0	31.7	44.5	7.4	130	0.689

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	69.9	60.5	66.6	59.3	64.0	58.2	62.8	66.5	61.1	64.1	59.7	62.4	8.4	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.3	83.0	64.1	59.9	78.9	62.3	58.5	75.1	60.7	28.6	66.6	27.5	64.0	26.5	62.5	72.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.0	24.6	21.7	DB	25.2	75.5	2.3	3.8	23.5	78.2	22.1	80.4	20.8	82.5	19.1	85.3	(4)
(5)			WB	24.5	66.6	2.4	2.7	22.7	68.5	21.3	70.0	20.0	71.5	18.2	73.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.8	39.8	40.4	42.4	45.8	50.3	54.4	56.9	56.6	54.0	48.9	43.6	40.5		
	DBStd	7.58	4.75	4.37	4.51	4.33	4.71	4.22	3.73	2.97	3.44	4.00	4.43	5.09		
	HDD50	1587	316	270	239	138	53	7	0	0	6	68	196	294		
	HDD65	6272	781	690	702	577	457	319	253	262	330	500	642	759		
	CDD50	797	0	0	2	11	62	139	215	204	126	33	4	1		
	CDD65	5	0	0	0	0	0	1	3	1	0	0	0	0		
	CDH74	27	0	0	0	0	0	5	17	5	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	10.7	12.0	11.8	11.4	10.7	10.9	9.8	9.1	9.6	10.1	10.7	10.6	11.6		
	PrecAvg	49.2	5.6	4.2	3.6	3.1	3.2	3.2	3.4	3.7	3.5	4.8	5.4	5.5		
(15) Precipitation	PrecMax	60.7	8.8	10.0	6.3	6.2	6.3	7.8	7.0	7.2	7.2	8.0	11.1	12.6		
	PrecMin	36.5	1.1	1.0	1.4	1.0	0.6	1.2	1.6	0.4	0.5	1.6	2.4	2.0		
	PrecStd	5.6	1.9	2.3	1.3	1.1	1.5	1.5	1.3	1.7	1.7	2.0	1.9	2.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.2	52.2	58.1	62.8	71.0	73.5	76.8	71.2	67.4	60.7	56.2	53.6		
		MCWB	51.1	50.3	50.1	52.9	59.1	61.6	66.8	63.1	60.8	57.5	55.2	52.6		
	2%	DB	50.3	50.6	53.0	58.3	66.1	68.9	69.7	66.1	64.0	58.0	53.7	51.5		
		MCWB	49.5	49.3	48.5	50.7	56.7	59.0	61.6	60.1	59.1	56.2	52.9	50.8		
	5%	DB	49.0	49.1	50.7	55.1	61.6	65.0	66.2	63.6	61.6	56.7	52.1	50.1		
		MCWB	48.3	48.1	47.9	49.3	53.9	57.7	60.2	59.0	57.8	55.0	51.2	49.5		
	10%	DB	47.5	47.6	49.1	52.6	58.2	61.8	63.0	61.7	59.7	55.1	50.5	48.4		
		MCWB	46.6	46.5	46.8	48.4	52.1	56.0	58.4	57.7	56.6	53.4	49.6	47.7		
	0.4%	WB	51.5	51.2	52.0	54.6	60.7	63.0	66.8	65.0	62.8	58.8	55.6	53.0		
		MCDB	52.0	51.6	54.6	60.2	68.6	70.3	74.6	69.6	65.2	59.5	56.0	53.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	49.9	49.8	50.2	52.4	57.4	60.6	63.2	62.1	60.7	57.0	53.1	51.0		
		MCDB	50.2	50.3	51.6	55.9	63.7	66.2	67.4	64.2	62.4	57.6	53.5	51.3		
	5%	WB	48.5	48.3	48.9	50.7	55.3	58.6	61.1	60.5	59.0	55.5	51.5	49.6		
		MCDB	48.9	48.9	50.2	53.6	59.7	63.1	64.1	62.2	60.6	56.2	51.9	50.0		
	10%	WB	46.8	46.8	47.2	49.1	53.3	56.9	59.3	59.0	57.4	53.9	49.8	47.8		
		MCDB	47.4	47.5	48.6	52.0	56.9	60.7	62.3	60.7	59.1	54.9	50.4	48.2		
	5% DB	MDBR	7.6	8.7	10.2	11.7	12.3	11.6	11.0	10.3	10.3	9.2	8.0	7.6		
		MCDBR	9.1	9.2	11.6	14.8	17.2	16.2	15.3	12.4	12.1	9.5	8.8	9.5		
		MCWBR	9.4	8.7	8.3	8.8	9.4	8.9	8.7	7.5	8.0	7.9	8.6	9.7		
		MCWBR	9.1	9.2	10.7	12.9	15.1	14.4	13.6	10.8	10.7	9.0	8.6	9.4		
(35) Mean Daily Temperature Range	5% WB	MCWBR	9.5	9.2	8.9	8.7	9.2	8.9	8.8	8.0	8.7	8.6	8.8	9.6		
	taub		0.329	0.338	0.357	0.369	0.379	0.364	0.367	0.352	0.349	0.334	0.325	0.318		
		taud	2.239	2.322	2.313	2.306	2.316	2.389	2.388	2.441	2.445	2.433	2.284	2.202		
(42) Clear-Sky Solar Irradiance	Ebn at Noon		192	238	259	272	275	280	277	276	261	238	193	168		
		Edh at Noon	20	25	31	36	38	36	35	31	28	23	19	17		
	RadAvg		196	414	768	1229	1548	1599	1418	1199	862	501	241	144		
(45) All-Sky Solar Radiation	RadStd		19	39	68	106	128	136	96	95	57	40	21	17		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (21 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page